Advanced Machine Learning, ETH Zurich

Training a neural network

One forward pass and one backward pass, whatever the network

Carlos Cotrini

November 2, 2026

Recap

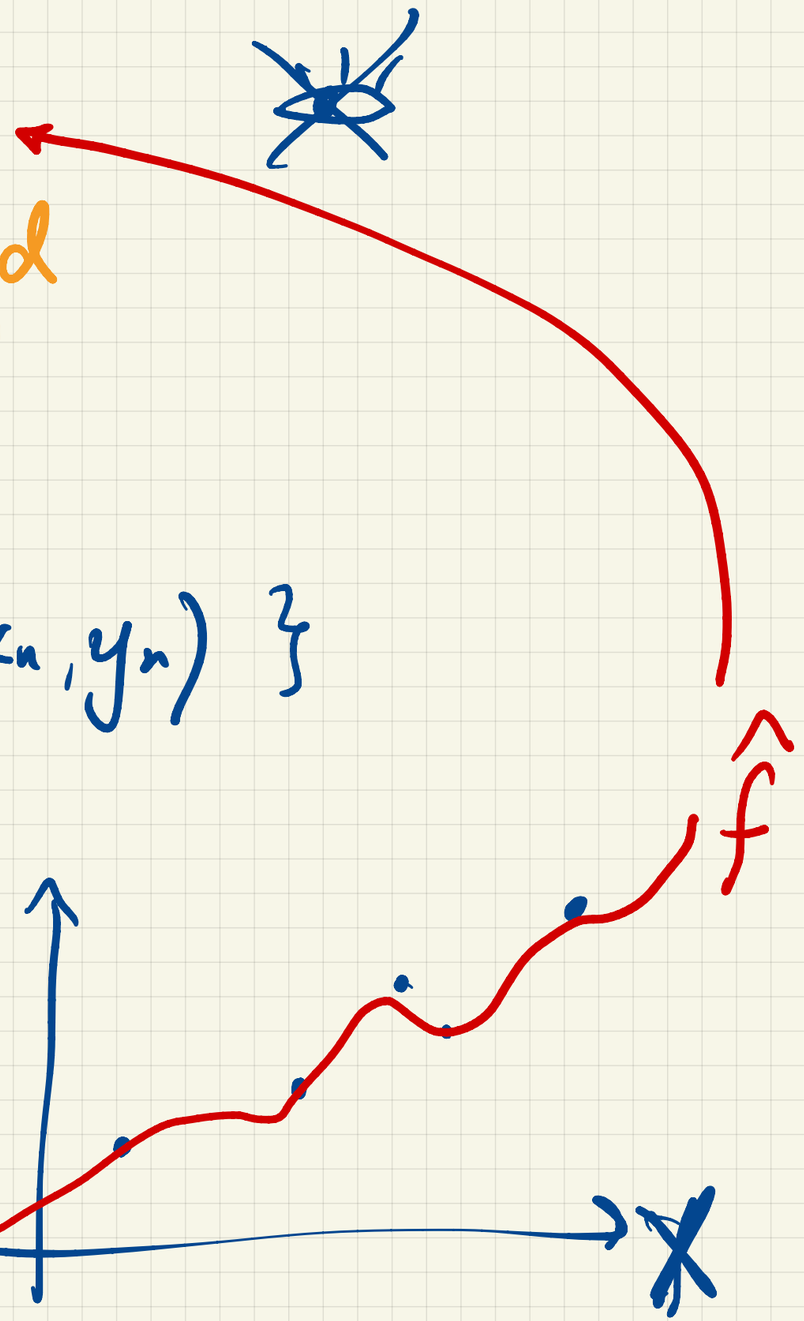
Hidden: $f : X \longrightarrow Y$ (price $\in \mathbb{R}$)

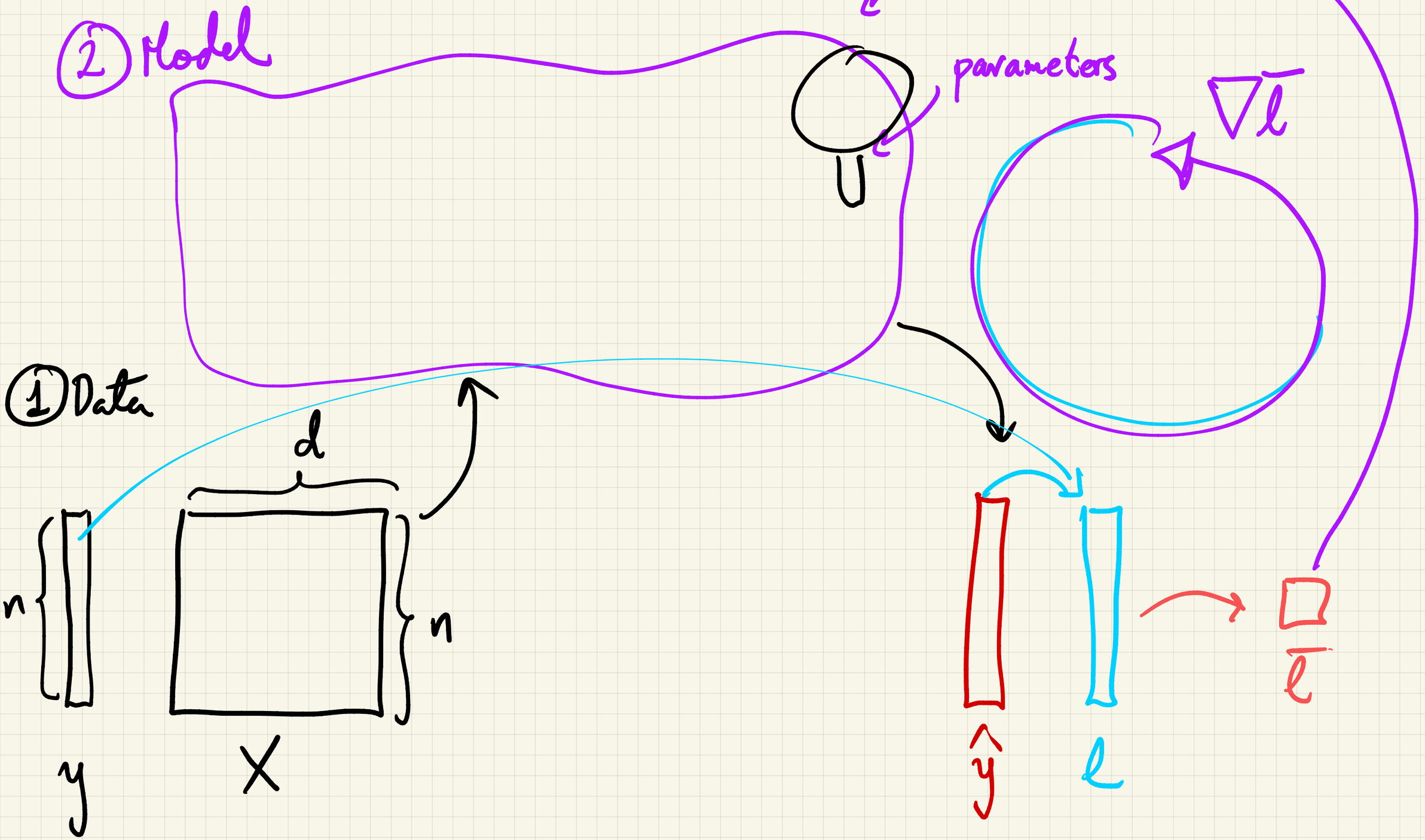
$X = (x_1, \ldots x_d) \in \mathbb{R}^d$

x_1: area, x_2: loc

Given: $\{ (x_1, y_1) \ldots \quad (x_n, y_n) \}$

n houses




2 Model
parameters
$\nabla \bar{l}$
1 Data
d
n
n
y
X
$\hat{y}$
l
$\bar{l}$

d

p

W

T

W

Layer 2

q

W'

p

q

1

W''

ϕ=ReLU

ϕ=ReLU

d

Layer 1

y

X

n

n

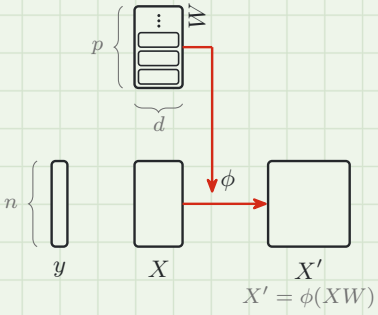
1

$X' = \phi(XW)$

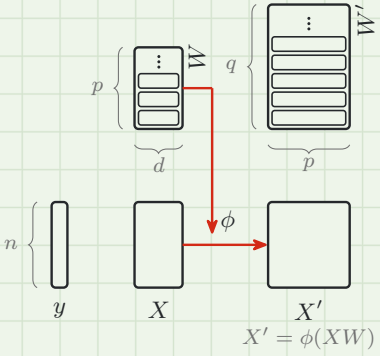
$X'' = \phi(X'W')$

$\hat{y} = \phi(X''W'')$

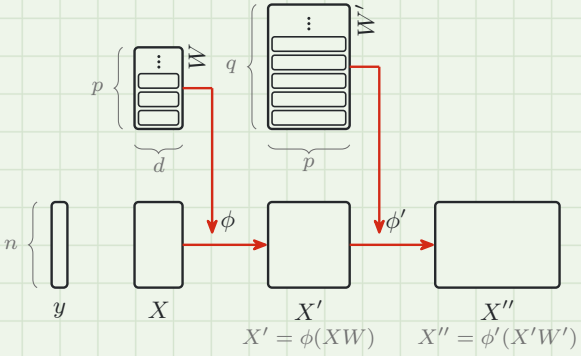
A network is a chain of blocks, each made from the one before by a block of weights and an activation ϕ like ReLU. The hinge fan has one hidden block, the hinge chain many.



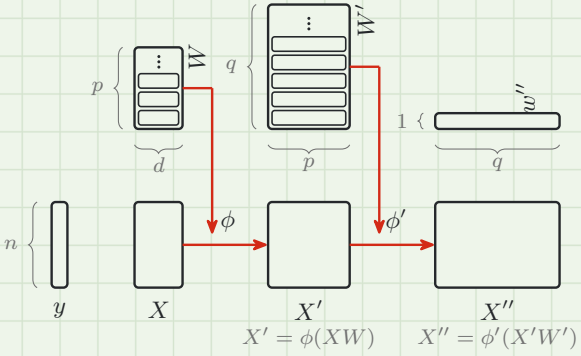
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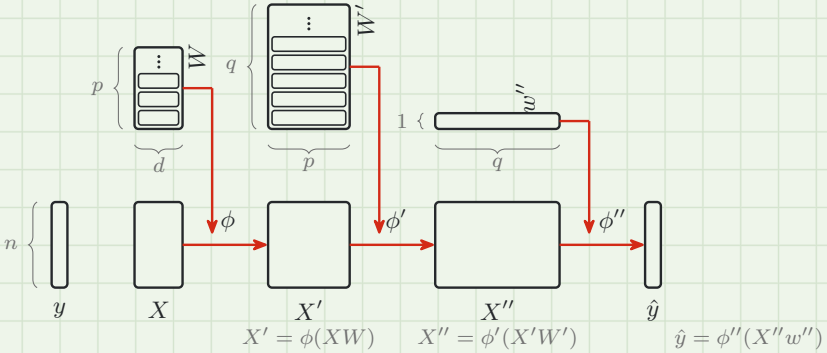
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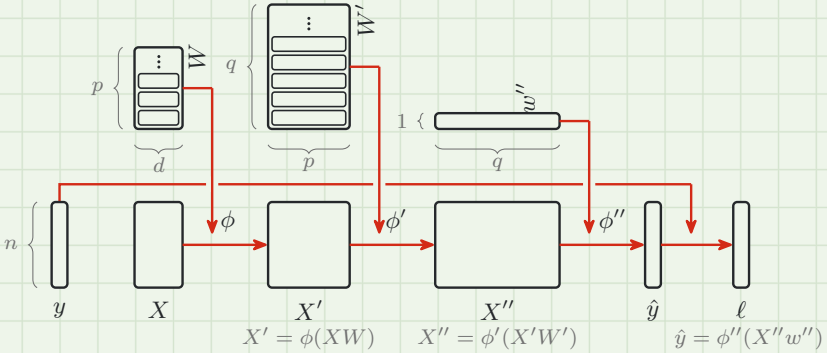
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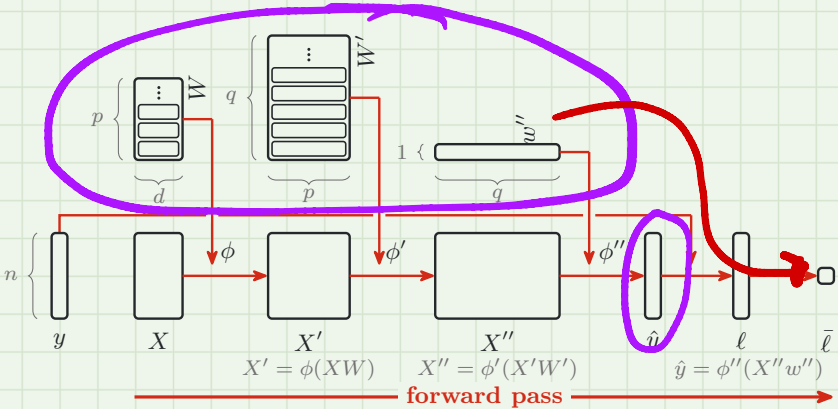
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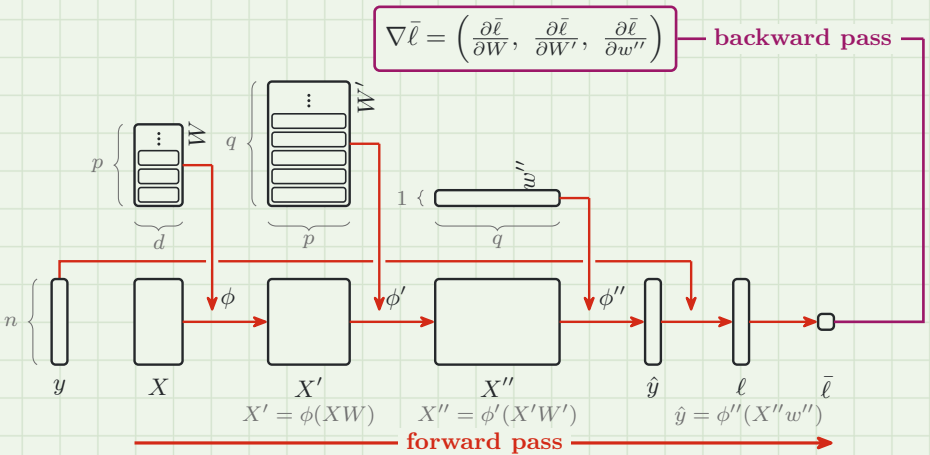
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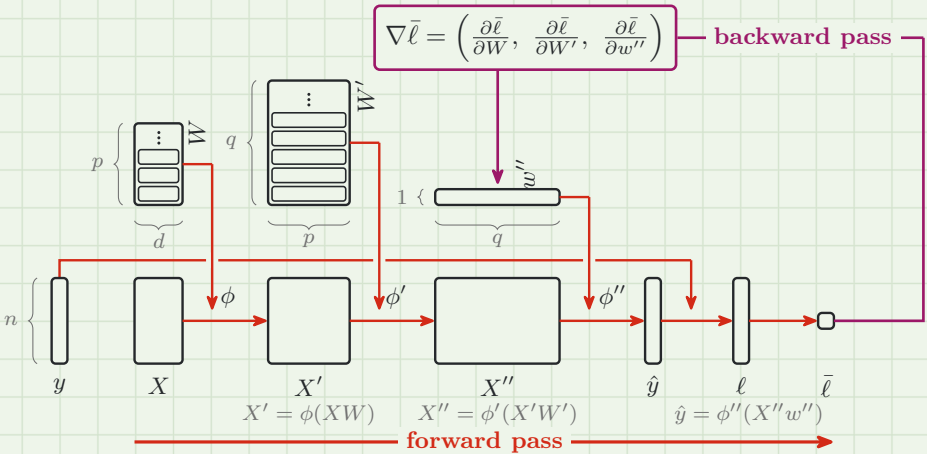
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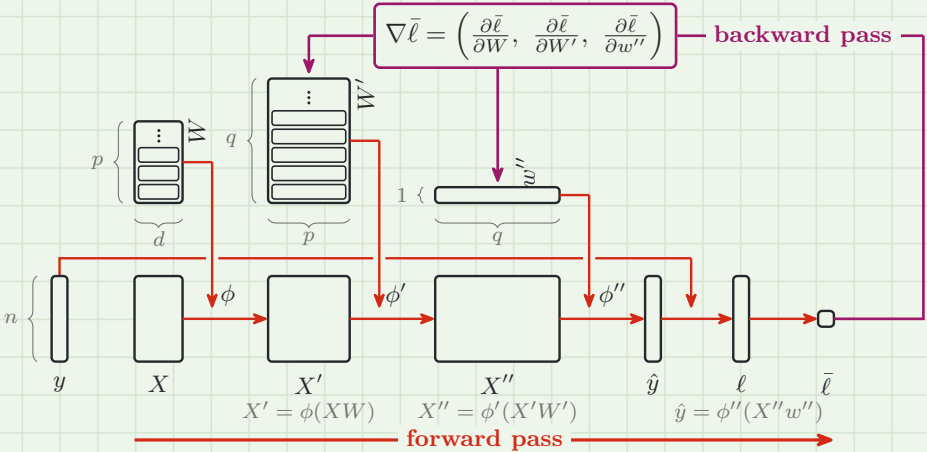
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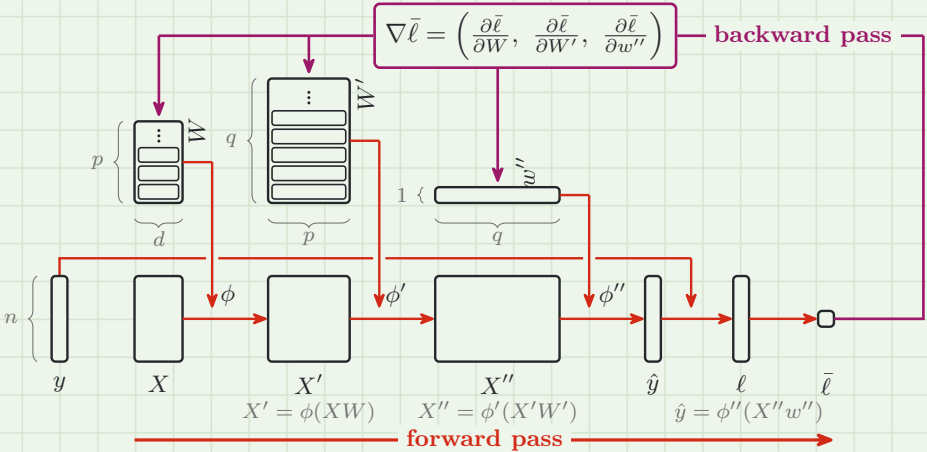
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